p-adic Analysis 1

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Problem 2

Let $x, y, z \in \mathbb{R}^n$. We wish to show that

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

If we redefine $u = x - y$ and $v = z - y$, this is equivalent to

$$\|u + v\| \leq \|u\| + \|v\|.$$

Squaring the inequality gives us

$$\|u + v\|^2 \leq \|u\|^2 + \|v\|^2 + 2\|u\|\|v\|.$$

We have that

$$\|u + v\|^2 = (u + v) \cdot (u + v) = \|u\|^2 + \|v^2\| + 2(u \cdot v),$$

and by the Cauchy-Schwarz inequality,

$$u \cdot v \leq \|u\|\|v\|,$$

$$\|u + v\|^2 = \|u\|^2 + \|v^2\| + 2(u \cdot v) \leq \|u\|^2 + \|v^2\| + 2\|u\|\|v\|,$$

as desired.

Problem 3

In $\mathbb{R}$, let

$$S_n = (-1/n, 1 + 1/n).$$

Then

$$\bigcap_{n \in \mathbb{N}} S_n = [0, 1],$$

which is closed.

Problem 6

Let $x, y, z \in \mathbb{R}$. We want prove the triangle inequality:

$$|e^x - e^y| + |e^y - e^z| \leq |e^x - e^z|.$$

Letting $a, b, c = e^x, e^y, e^z$, this is equivalent to the regular triangle inequality on the standard metric on $\mathbb{R}$. This metric space is not complete. Take $x_n = -n$. This is Cauchy since $e^{-n} \to 0$, but there is no $x \in \mathbb{R}$ such that $e^x = 0$. The completion would be $[0, \infty]$, adding 0 as the only limit of the Cauchy sequences that don't converge.

Problem 7

Let $\varepsilon > 0$. There exists N such that if $i > N$ then

$$d(a_i, b_i) < \varepsilon/2, \quad d(b_i, c_i) < \varepsilon/2.$$

By triangle inequality,

$$d(a_i, c_i) \leq d(a_i, b_i) + d(b_i, c_i) < 2 \cdot \varepsilon/2 = \varepsilon,$$

so a is equivalent to c.

Problem 8

Since

$$\lim_{p\to\infty} d_p(x, y) = \lim_{p\to\infty} \left(\sum (y_i - x_i)^p\right)^{1/p} = \max_i (y_i - x_i),$$

$d_\infty(x, y)$ is the maximum difference between any two individual coordinates of x, y.

Problem 9

Let $x, y \in X$ and $d = \|y - x\|$. Let $U = B_{d/2}(x), V = B_{d/2}(y)$. Then $U \cap V = \varnothing$ because, if a point z is $< d/2$ away from x, by triangle inequality, it must be $> d/2$ away from y.

Problem 12

Assume a Cauchy sequence a_n converges to x and y with $x \neq y$. Let $0 < \varepsilon < d(x, y)/2$. We have that $B_\varepsilon(x) \cap B_\varepsilon(y) = \varnothing$, but since a_n converges to x and y, there exists N such that if $n > N$, $a_n \in B_\varepsilon(x)$ and $B_\varepsilon(y)$, which is a contradiction. So $x = y$.

Problem 13

Consider the sequence

$$x_n = \sum_{k=0}^{n} p^k.$$

If $m > n$ then

$$x_m - x_n = p^{n+1}(1 + p + \cdots + p^{m-n-1},$$

so

$$d_p(x_m - x_n) = p^{-(n+1)} \to 0 \text{ as } n \to \infty.$$

So x_n is Cauchy in $\mathbb{Q}$, but there is no $x = \sum_{k=0}^{\infty} p^k$ in $\mathbb{Q}$, so the metric space is not complete.

Problem 16

Assume $f : (X, d_X) \to (Y, d_Y)$ satisfies the ε-δ definition. Take any open $U \subseteq Y$. We want to show that $f^{-1}(U)$ is open in X.

Pick any $x_0 \in f^{-1}(U)$, so that $f(x_0) \in U$. Since U is open, there exists $\varepsilon > 0$ such that

$$B_Y(f(x_0), \varepsilon) \subseteq U.$$

Since f is continuous at x_0, there exists $\delta > 0$ such that

$$d_X(x_0, x) < \delta \implies d_Y(f(x_0), f(x)) < \varepsilon.$$

So if $x \in B_X(x_0, \delta)$, then $f(x) \in B_Y(f(x_0), \varepsilon) \subseteq U$, so $x \in f^{-1}(U)$. Therefore, $B_x(x_0, \delta) \in f^{-1}(U)$, and $f^{-1}(U)$ is open.

Now assume that for every open $U \subseteq Y$, $f^{-1}(U)$ is open in X. Fix $x_0 \in X$ and $\varepsilon > 0$. Let

$$U = B_Y(f(x_0), \varepsilon).$$

Since U is open, $f^{-1}(U)$ is open in X, and also $x_0 \in f^{-1}(U)$. So there exists $\delta > 0$ such that

$$B_X(x_0, \delta) \subseteq f^{-1}(U).$$

Therefore, whenever $d_X(x_0, x) < \delta$, $x \in f^{-1}(U)$, so

$$f(x) \in U = B_Y(f(x_0), \varepsilon)$$

so $d_Y(f(x_0), f(x)) < \varepsilon$.